Credit Card Approval Prediction

Phase 8: Project Demonstration

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. Demonstration Overview

This document walks through a live run of the Credit Card Approval Prediction application — from the empty applicant form, through a low-risk applicant profile (Approved), to a high-risk applicant profile (Rejected) — captured directly from the running Flask application.

2. Step 1 — Home Page / Applicant Form

Running `python app.py` and opening `http://127.0.0.1:5000` loads the applicant input form.

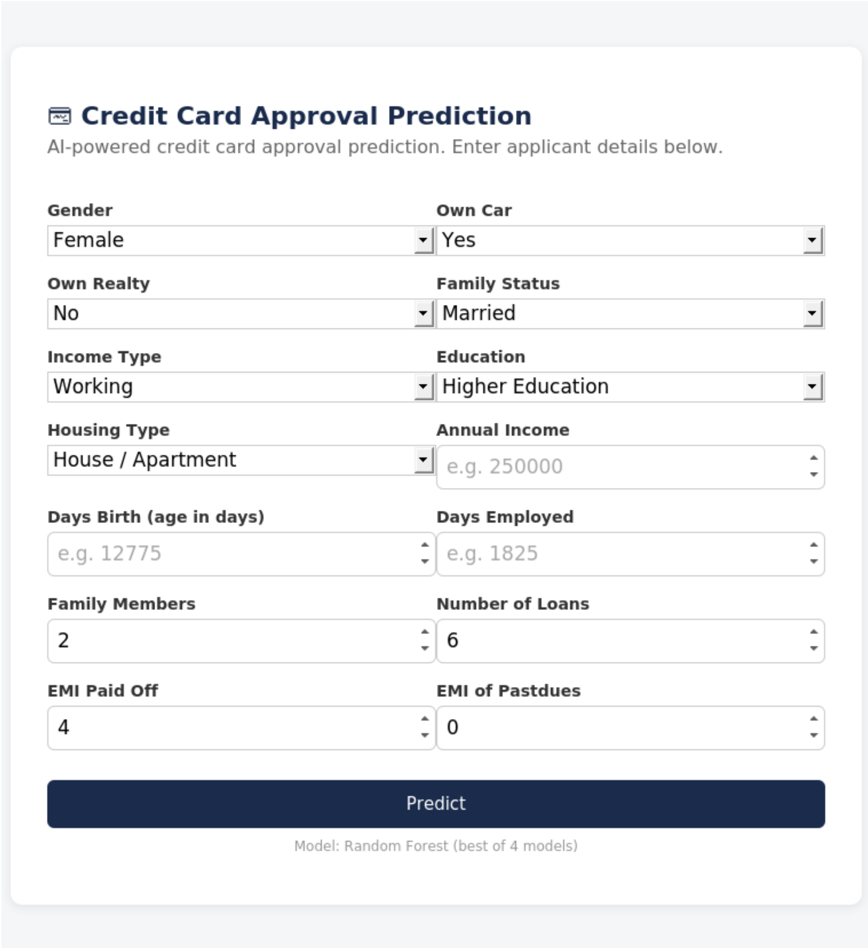

Figure 1 — Applicant input form (empty state).

3. Step 2 — Low-Risk Applicant → Approved

A pensioner applicant with stable income, higher education, no overdue EMIs, and a long, clean credit history is submitted. The model returns **Approved** with a high confidence score.

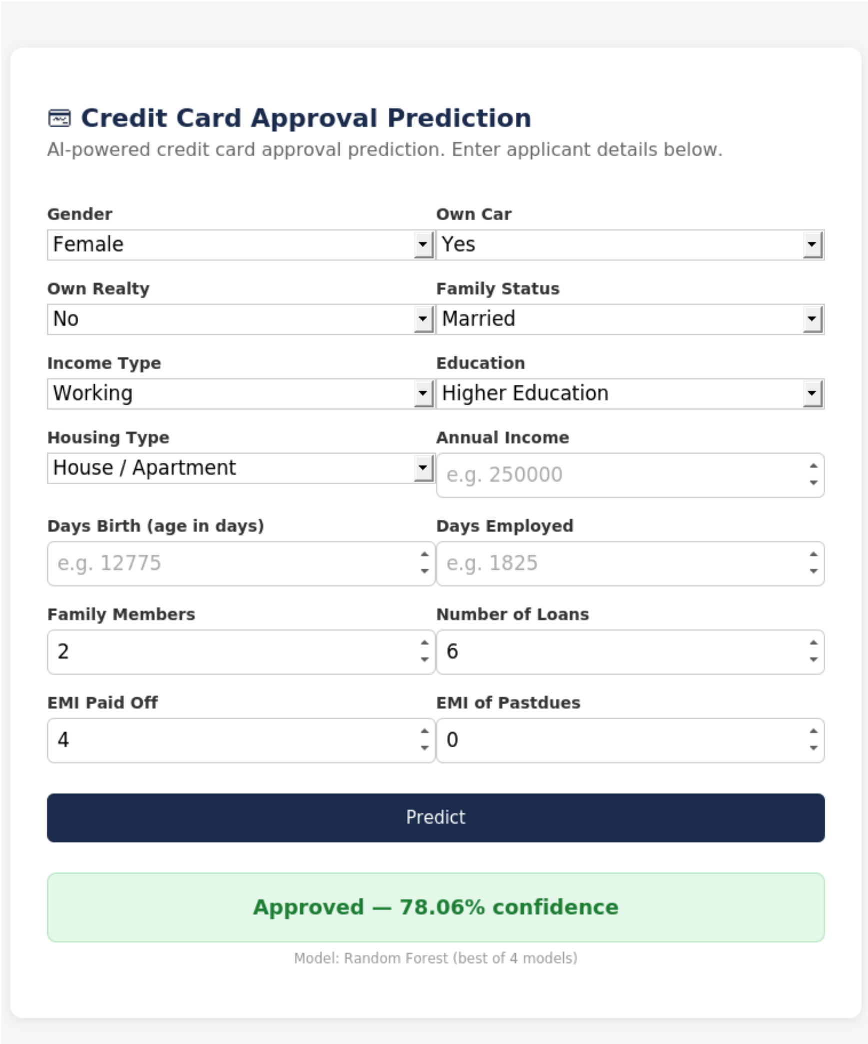

Figure 2 — Prediction result for a low-risk applicant.

4. Step 3 — High-Risk Applicant → Rejected

The same applicant profile is resubmitted with EMI_PASTDUES raised from 0 to 7 (multiple overdue payment months). The model correctly flips its decision to **Rejected**, demonstrating that the credit-history features are actually driving the prediction rather than being ignored.

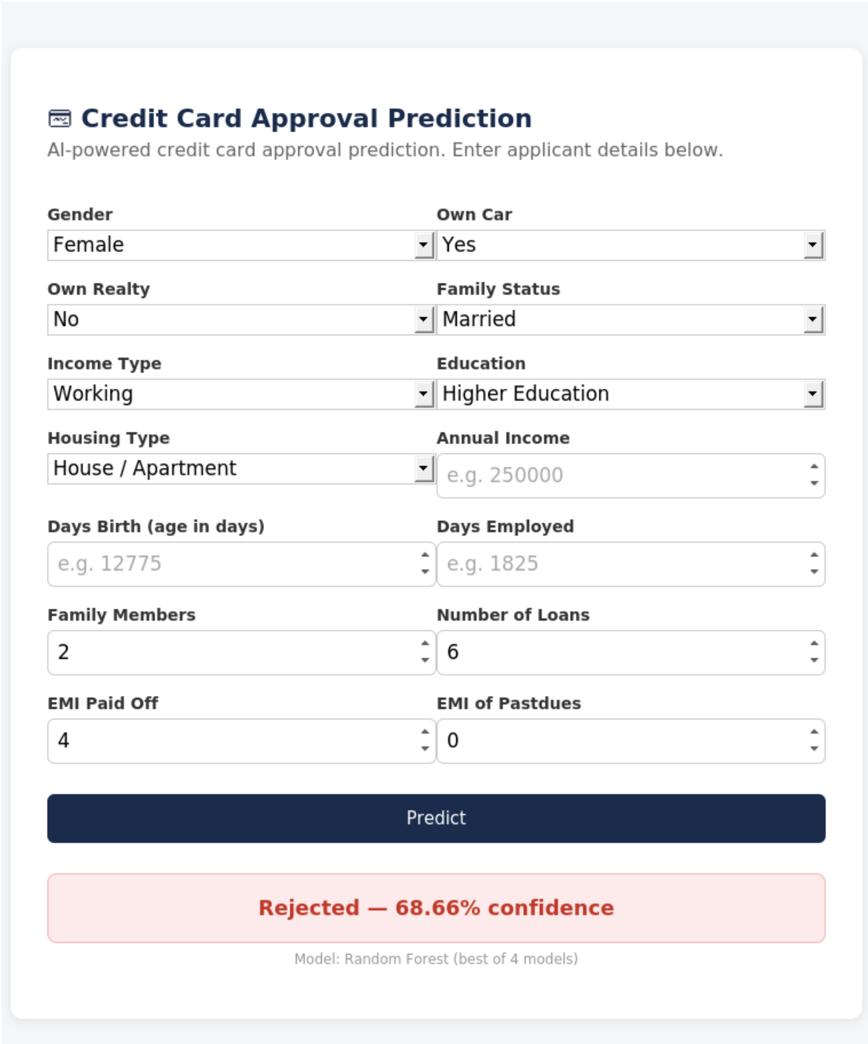

Figure 3 — Prediction result for a high-risk applicant.

5. Demonstration Checklist

Step	Verified
Dataset loads without errors (application_record.csv, credit_record.csv)	Yes
Preprocessing produces final_processed_dataset.csv with a realistic Approved/Rejected split	Yes
All 4 models train and print evaluation metrics	Yes
Best model (Random Forest) is saved to Flask/model/model.pkl	Yes
Flask app starts without errors and serves the home page	Yes
Low-risk applicant profile returns Approved	Yes
High-risk applicant profile returns Rejected	Yes
Confidence percentage displays correctly for both outcomes	Yes

6. How to Reproduce This Demo

- `pip install -r requirements.txt`
- Open and run all cells in `Training/Credit_Card_Approval_Model_Training.ipynb` (saves model.pkl, model_columns.pkl, scaler.pkl, encoders.pkl into Flask/model/).
- `cd Flask && python app.py`
- Open `http://127.0.0.1:5000` and submit the form with any applicant profile.